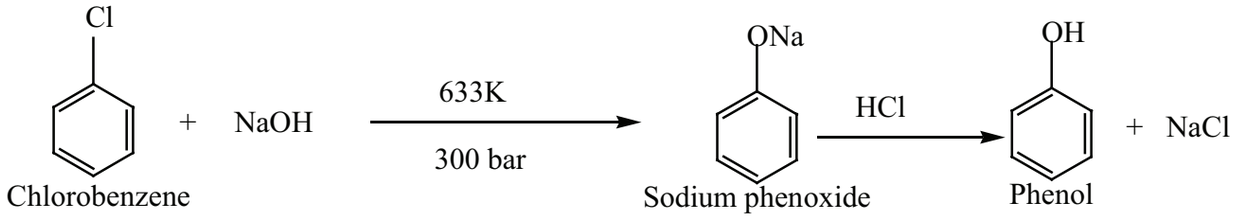
Preparation of Phenols

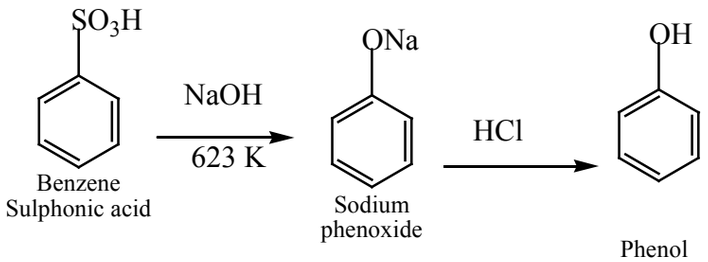
a) From halo arenes(Dows process)

When Chlorobenzene is hydrolysed with 6-8% NaOH at 300 bar and 633K in a closed vessel,sodium phenoxide is formed which on treatment with dilute HCl gives phenol.



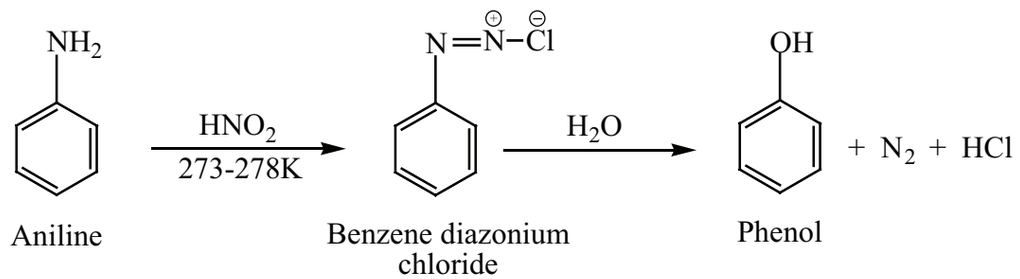
b) From benzene sulphonic acid

Benzene is sulphonated with oleum and the benzene sulphonic acid so formed is heated with molten NaOH at 623K gives sodium phenoxide which on acidification gives phenol.



c) From aniline

Aniline is diazotized with nitrous acid ($NaNO_2$+HCl) at 273-278K to give benzene diazonium chloride which on further treatment with hot water in the presence of mineral acid gives phenol.



d) From cumene

A mixture of benzene and propene is heated at 523K in a closed vessel in presence of H_3PO_4 catalyst gives cumene (isopropylbenzene). On passing air to a mixture of cumene and 5% aqueous sodium carbonate solution, cumene hydro peroxide is formed by oxidation. It is treated with dilute acid to get phenol and acetone. Acetone is also an important byproduct in this reaction.

XII U11-Hydroxy compounds.indd 126 16-12-2022 19:42:11